1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), tumors that are TLS-rich by histology but harbor very high TMB and deficient mismatch repair (dMMR) exhibit dysfunctional TLS maturation and CD8+ T-cell exhaustion driven by extensive neoantigen heterogeneity, leading to suboptimal response to immune checkpoint inhibitors (ICIs) compared with TLS-rich tumors with lower heterogeneity or TLS-poor tumors with similar TMB/dMMR. Therefore, TLS presence alone is insufficient as a predictor of ICI response in the TMB-high/dMMR setting; integrated biomarkers that combine (a) TLS maturation status (e.g., presence of germinal centers, organized B/T compartments), (b) clonal vs subclonal neoantigen burden (neoantigen clonality/heterogeneity), and (c) MMR/TMB status will better stratify patients for immunotherapy. This model yields testable predictions: (1) TLS maturation scores will correlate with ICI responses primarily in tumors with high clonal neoantigen burden; (2) among TLS-rich, TMB-high/dMMR tumors, those with a high fraction of subclonal neoantigens will show reduced objective response rates and shorter progression-free survival due to TLS dysfunction and T-cell exhaustion; (3) patients with discordant TLS maturation and neoantigen clonality will benefit from combination strategies targeting TLS restoration or T-cell exhaustion pathways. Secondary modifiers (TP53, DPYD, GARS1, LRPPRC) may modulate these relationships by influencing tumor biology, metabolism, and immune contexture, but the primary predictive signal arises from the interaction of TLS architecture, TMB/dMMR, and neoantigen clonality.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TMB, dMMR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TMB: TMB/dMMR) and display organized immune structures nevertheless fail to benefit from immunotherapy — why does apparent immune organization (TLS/TILs) not uniformly translate into effective checkpoint responsiveness? 2) Tier 1 paths used: I built on the dMMR/MSI → ICI responsiveness path (mechanistic neoantigen/repair-defect axis) and the TMB → immunotherapy biomarker path (quantitative mutational burden driving neoantigen load), and I incorporated the BCG/biomarker context conceptually to motivate local immune-organizing structures; these jointly suggest a tension between mutational load-driven immunogenicity and local immune function.

dMMR: dMMR) and display organized immune structures nevertheless fail to benefit from immunotherapy — why does apparent immune organization (TLS/TILs) not uniformly translate into effective checkpoint responsiveness? 2) Tier 1 paths used: I built on the dMMR/MSI → ICI responsiveness path (mechanistic neoantigen/repair-defect axis) and the TMB → immunotherapy biomarker path (quantitative mutational burden driving neoantigen load), and I incorporated the BCG/biomarker context conceptually to motivate local immune-organizing structures; these jointly suggest a tension between mutational load-driven immunogenicity and local immune function.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TP53	Supporting modifier from knowledge graph
Secondary	DPYD	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	LRPPRC	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: on CD8+ T cells, reducing cytotoxic function. A biomarker model that jointly considers TLS architecture, neoantigen clonality (clonal vs subclonal burden), and TMB/dMMR status should better predict ICI outcomes and identify patients who may need TLS-restoring or anti-exhaustion combination therapies.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- DPYD: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.